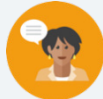


3.6 Fitting Linear Functions to Data

Lesson Introduction

 Benchmark	MA.912.DP.2.4: Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y-intercept of the model. Use the model to solve real-world problems in terms of the context of the data.		 Learning Targets	- I can formally fit a linear function to bivariate numerical data using technology. - I can relate the correlation coefficient to the slope and fit of a linear relationship between two variables.	
 Benchmark Coherence	Building On	Working Toward			
	MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.	MA.912.DP.2.8 Fit a quadratic function to bivariate numerical data that suggests a quadratic association, and interpret any intercepts or the vertex of the model. Use the model to solve real-world problems in terms of the context of the data. MA.912.DP.2.9 Fit an exponential function to bivariate numerical data that suggests an exponential association. Use the model to solve real-world problems in terms of the context of the data.			
 Essential Questions	- How do you use technology to fit a linear function to bivariate numerical data? - How can you relate the correlation coefficient to the slope and fit of a linear relationship between two variables?				
 Lesson Narrative	In this lesson, students learn how to use technology to fit a linear function to bivariate numerical data. Students will learn to relate the correlation coefficient to the slope and fit of a linear relationship between two variables. Students will also begin using lines fit to data to solve real-world problems.				
 Component Summary	3.6.1 Warm-Up (~5 min.)	Notice and Wonder scatter plot activity (<i>SE p. 138</i>)	3.6.3 Guided Instruction (10 min.)	Learn to relate the correlation coefficient to the slope of the fit of the regression line (<i>SE p. 140</i>)	
	3.6.2 Exploration (10-15 min.)	Explore formally fitting a linear function to bivariate data using technology (<i>SE p. 138</i>)	3.6.4 Guided Practice (10-15 min.)	Practice formally fitting a line of best fit to data, and using it to solve problems (<i>SE p. 142</i>)	
	3.6.5 Wrap-Up (~5 min.)	Complete the statements reflection activity (<i>SE p. 143</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> Correlation Coefficient <u>Additional Glossary:</u> N/A		(Optional) Graphing utility for the data in 3.6.2 Exploration		(Optional) Anchor chart for correlation coefficient

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may always say “strong” or “weak” in all cases, and not use words such as “moderately” for less strong or weak patterns of association. - Students may confuse positive and negative associations.	Students will be exposed to visuals of statistical data on a calculator screen. They may need assistance with understanding what each piece means and why it is there, as it will continue to reappear for different contexts throughout this course and beyond.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 3.6.1 Warm-Up provides students with a scatter plot and has them identify things that they notice and things that they wonder. The purpose of this instructional routine is to make a mathematical task accessible for all students with these two, low-stakes questions.	MA.K12.MTR.7.1: Real-World Contexts In 3.6.4 Guided Practice, students will apply what they have learned in the prior activities to a real-world scenario. This allows students to connect the mathematical concepts of this lesson to everyday experiences. They will use models to understand and solve real-world problems in context.
 Differentiate Instruction	Consider using technology in this lesson. Provide students with different technology options, such as graphing calculators or online graphing software, to give them a range of options for exploration during this lesson.	
Lesson Notes:		

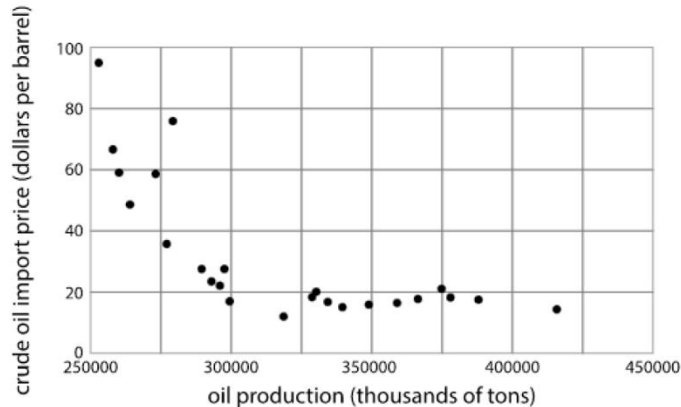
3.6.1 Warm-Up: Notice and Wonder

Component Overview: Students will consider a scatter plot to notate one thing that they notice and one thing that they wonder.



3.6.1 Warm-Up: Notice and Wonder

The scatter plot shows the relationship between oil production, in thousands of tons, and the crude oil import price, in dollars per barrel.



This activity provides students with a scatter plot and has them identify things that they notice and things that they wonder. The purpose of this instructional routine is to make a mathematical task accessible for all students with these two, low-stakes questions. It also provides the opportunity to make connections to data collected for real-world contexts. Their notices and wonderings could be mathematical and/or relate to the context itself.

1. Complete the table to indicate one thing you notice and one thing you wonder about when looking at this scatter plot.

One Thing I Notice is ...	One Thing I Wonder is ...
Answers will vary. I notice that as production increases the price per barrel tends to decrease.	Answers will vary. I wonder why there isn't an increase in oil production so the cost stays lower for everyone.

3.6.2 Exploration: Formally Fit a Linear Function to Bivariate Data Using Technology

Component Overview: Students will be provided a set of data to explore the process for formally fitting a linear function to bivariate data using technology.



3.6.2 Exploration: Formally Fit a Linear Function to Bivariate Data Using Technology

The mass, in kilograms (kg), and fuel efficiency, in miles per gallon (mpg), of 20 new gas-powered cars is given in the table.

Mass (kg)	Fuel Efficiency (mpg)
1549	25
1610	20
1737	21
1777	20
1155	28
1486	23
1962	16
2384	16
1957	19
2212	16
1115	29
2068	18
1663	19
2068	18
1663	19
2216	18
1432	25
1987	18
1259	25
1580	26
1234	30
1656	23

The raw data can be used to formally fit a line to model the relationship between the data using technology.



This activity shows, through images, how technology can be used to formally fit a line. If the technology is available in the classroom, consider using the link in the digital teacher area to go to the actual graphing utility with this data entered to explore the linear regression line.



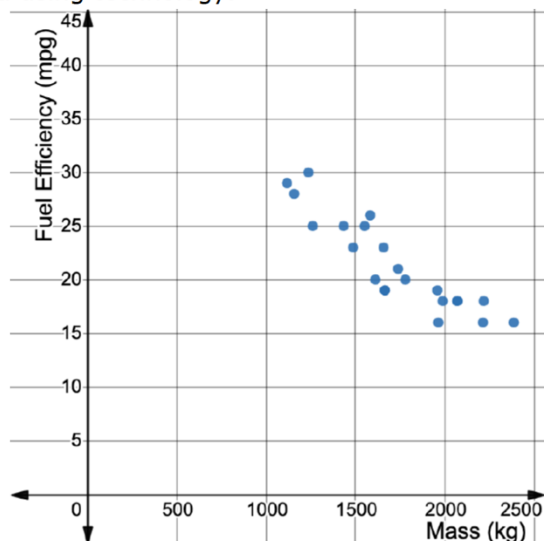
This exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share with the whole group.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

3.6.2 Exploration: Formally Fit a Linear Function to Bivariate Data Using Technology (continued)

1. Tess created the scatter plot shown to represent the data using technology.



Work with your partner to describe the pattern of association between a car's mass and fuel efficiency.

Strong, negative, linear association. As a car's mass increases, the fuel efficiency decreases.

2. Tess fit a line to the data using a graphing calculator. The screenshot shows the results of Tess generating a line, in slope-intercept form, to model the relationship between the mass of a car, x_1 , and the car's fuel efficiency, y_1 .

$$y_1 \sim mx_1 + b$$

STATISTICS	RESIDUALS
$r^2 = 0.8235$	e_1 plot
$r = -0.9075$	
PARAMETERS	
$m = -0.0109749$	$b = 40.2965$



Consider having students try to recreate this image using their graphing technology as a method to practice the continued effective use of their graphing utility.



To have students further analyze what they are looking at and prompt interest into what the “ r ” represents, consider the following questions:

- What do you notice about the information in this image?
- What are some things that you wonder?

Note that the meaning of “ r ” will be introduced in the next component, so it is not necessary students understand what this is at this point.



3.6.2 Exploration: Formally Fit a Linear Function to Bivariate Data Using Technology (continued)

- a. The “~” symbol is used to mean one quantity is “approximately equivalent to” another quantity. Discuss with your partner why you think this symbol would be used rather than an “=” when determining a line of best fit. Summarize your discussion.

Answers will vary. A line of best fit is used to estimate the relationship between the two variables – the line is an approximation of the relationship and can be used to make predictions.

- b. According to the screenshot, what are the slope and y-intercept of the line of best fit generated using technology, rounded to the nearest thousandth?

slope	−0.011
y-intercept	40.297

- c. Write the slope-intercept form of the equation of the line of best fit Tess generated with technology.

$$y = -0.011x + 40.297$$

- d. Interpret the slope in the given context.

The fuel efficiency of a car decreases 0.011 miles per gallon, as the mass increases by 1 kg.

- e. Interpret the y-intercept in the given context.

A car that weighs 0 kg would be expected to have a fuel efficiency of 40.297 mpg.



Why would the “~” symbol not be used in earlier scenarios, such as graphing linear functions?



Listen for student responses to parts d and e. A common misconception is for students to leave out essential words and/or phrases, such as “as the mass increases by 1 kg” in question 2 part d.

3.6.2 Exploration: Formally Fit a Linear Function to Bivariate Data Using Technology (continued)

- f. Explain whether the y -intercept makes sense in this context.

It does not make sense because a car cannot weigh 0 kg.

Extrapolation occurs when a model is used to make predictions for x -values that fall outside of the data set.

Problems may occur when extrapolating, since extending the trend line far past the x -values of the data set may make the predictions less reliable.

3. Explain why it would not be reliable to use Tess' model for this data set to make predictions about the fuel efficiency of hybrid or electric vehicles.

Answers will vary. It would not make sense because they have additional/different design features that cause them to get additional mpg outside of traditional weight factors.

4. Use inequalities to express an appropriate domain and range for making predictions using the linear equation modeling the data.

domain	$1000 \leq x \leq 2500$
range	$15 \leq y \leq 31$



Challenge students to see if they can think of a real-world scenario where data has been extrapolated to support a position.



Why is the domain not all real numbers in this scenario?

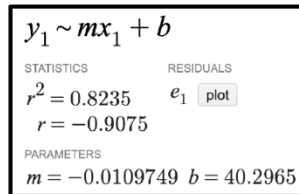
3.6.3 Guided Instruction: Relating the Correlation Coefficient, r , to the Slope of the Fit of the Regression Line

Component Overview: Students will be introduced to the correlation coefficient and how to relate it to the slope of the fit of the regression line.



3.6.3 Guided Instruction: Relating the Correlation Coefficient, r , to the Slope of the Fit of the Regression Line

The screenshot of the line Tess formally fit to the data using technology included a statistic, r .



The statistic, r , is the **correlation coefficient**. It measures the strength and direction of the linear association between the variables in a set of bivariate numerical data.



When launching this activity, do a call back quiz to the image from the calculator. Ask students to share what information they can gather from that screen, and what things from the image they are still unsure about what it represents.



The **correlation coefficient**, r , is a unitless value between -1 and 1 .

- A correlation coefficient of 1 represents a perfect linear relationship with a positive direction.
- A correlation coefficient of -1 represents a perfect linear relationship with a negative direction.
- The closer the correlation coefficient is to 0 , the weaker the linear association between the variables.



The information about the correlation coefficient would make a great anchor chart. Students can use this information as a point of reference as they look at associations of data, both in this unit and beyond.

3.6.3 Guided Instruction: Relating the Correlation Coefficient, r , to the Slope of the Fit of the Regression Line (continued)

1. Complete the statement to interpret the correlation coefficient for Tess' linear model.

The correlation coefficient -0.9075 means there is a

☐ weak
☒ strong

☒ linear
☐ nonlinear

association between a car's

mass and its fuel efficiency because the value of r is

close to

☐ 0.
☒ -1 or 1 .

The sign of r is

☐ positive,
☒ negative,

which means the direction of the association between

the variables is

☐ positive.
☒ negative.

2. Albert, an ice cream vendor at Jones Beach, records the number of cones he sells each day, as well as the daily high temperature. The table and the scatter plot below show his data for 12 days in June.

Relationship Between Temperature and Cones Sold

Temperature ($^{\circ}F$)	81	72	88	85	89	90
Number of Cones Sold	55	36	67	65	72	75
Temperature ($^{\circ}F$)	87	75	83	79	87	93
Number of Cones Sold	73	40	60	45	69	72

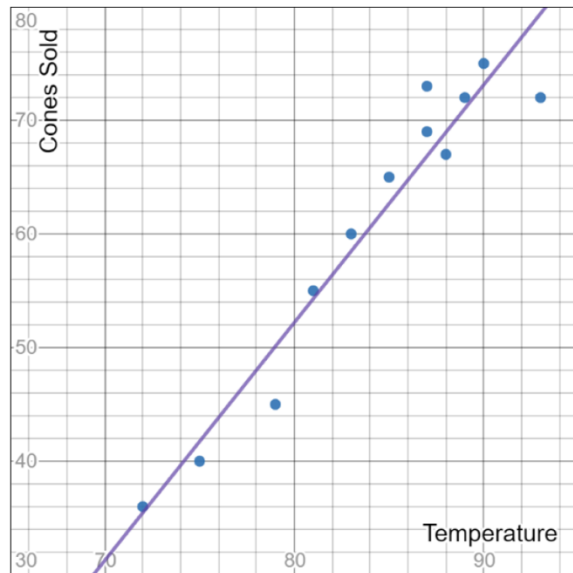


Model question 1 for students to show how the information provided earlier is applied to a specific scenario. Include the use of thinking out loud when deciding which word to choose so that students can follow along and model a similar thinking pattern for future contexts.



For practice, if time permits, have students graph this data using graphing technology.

3.6.3 Guided Instruction: Relating the Correlation Coefficient, r , to the Slope of the Fit of the Regression Line (continued)



The equation of the line of best fit for this data is $y = 2.0925x - 115.195$ with a correlation coefficient $r = 0.9646$.

Complete the statements.

There is a ☐ weak, ☒ strong, ☐ positive, ☐ negative, ☐ linear, ☐ nonlinear association between the temperature outside and the number of ice cream cones Albert sells in June. The number of cones he sells ☐ increases ☐ decreases by approximately ☐ 2 ☐ 115 for every 1°F increase in the temperature outside.



To help prepare students to complete the statement, consider asking the following questions:

- What are some things that you notice about this data display?
- What are some things that you wonder?



Consider leveraging a Think-Pair-Share instructional routine on this question. Allow students a minute or two to try it on their own. Then, have them pair up with a partner to share their answers and resolve any differences to come to a consensus.

3.6.4 Guided Practice: Formally Fitting a Line of Best Fit to Data and Using it to Solve Problems

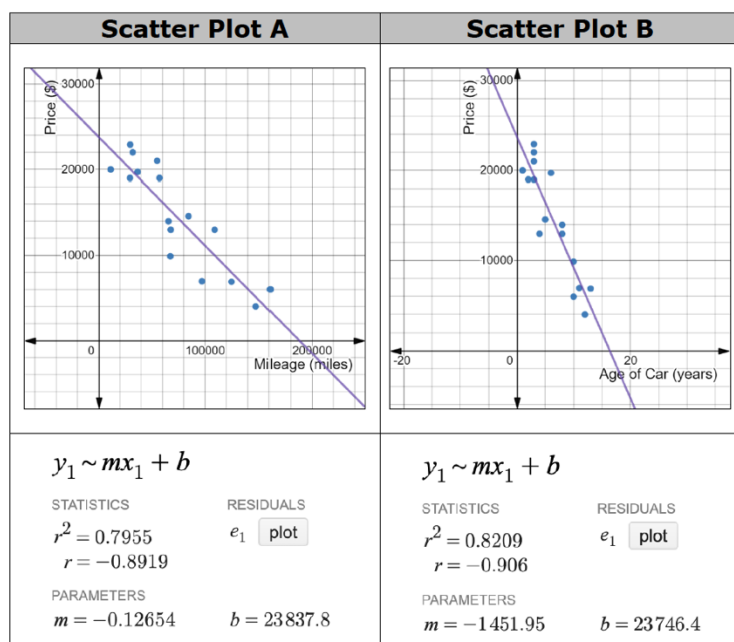
Component Overview: Students will practice fitting a line of best fit to data and using it to solve a real-world problem.



3.6.4 Guided Practice: Formally Fitting a Line of Best Fit to Data and Using it to Solve Problems

- Kara wants to sell her Subaru Forester, but does not know what the listing price should be. She checks on a popular used-car site and finds 16 Subaru Foresters listed. Kara uses the data to construct two different scatter plots – one that relates the vehicle's age, in years, to the sale price, and the other that relates the number of miles on each vehicle to its sale price.

Two different scatter plots with lines of best fit that were generated using technology are shown. Scatter Plot A relates the mileage of each vehicle to the sale price. Scatter Plot B relates the age of each vehicle to the sale price.



In this activity, students will apply what they have learned in the prior activities to a real-world scenario. This allows students to connect the mathematical concepts of this lesson to everyday experiences.



What are some of the similarities that you notice between the two scatter plots? What are some of the differences that you notice?

3.6.4 Guided Practice: Formally Fitting a Line of Best Fit to Data and Using it to Solve Problems (continued)

- a. Write the equation for each line of best fit using the values generated with technology.

Line of Best Fit for Scatter Plot A	Line of Best Fit for Scatter Plot B
$y = -0.12654x + 23,837.8$	$y = -1451.95x + 23746.4$

- b. Interpret the slope of the line of best fit for Scatter Plot B.

The sale price of a Subaru forester decreases by \$1,451.95 as it ages each year.

- c. Describe the strength and direction of the linear association between mileage and sale price of a used Subaru Forester.

There is a strong, negative, linear association between these variables.



Listen for student responses to question 1 part b. A common misconception is for students to leave out essential words and/or phrases, such as “as it ages each year.”

In question 1 part c, ensure that students are using all the appropriate language for describing the association. A common error is for students to say strong linear or negative linear and leave out the other necessary descriptive language.

2. Suppose Kara’s Subaru Forester is 9 years old.

- a. Which equation could she use to predict a good sale price for her car?

$$y = -1451.95x + 23,746.4$$

- b. What sale price should Kara list her car for based on its age?

\$10,678.85



Suppose Kara’s Subaru has 95,000 miles on it. Discuss with your partner a good sale price based on both the age of her car, and the number of miles on it.

3.6.5 Wrap-Up: Reflection

Component Overview: Students will complete two statements about the concepts they remain unsure about after the lesson, and how they intend to fill in that gap.



3.6.5 Wrap-Up: Reflection

1. Complete each statement.

a. After today's lesson, I am still unsure about ...

Answers will vary.

b. To fill in this gap, I intend to ...

Answers will vary.



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.